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Chapter 1

Viscoelastic Fluid Mechanics and the Four-Roll Mill

In this chapter, the continuum mechanics framework governing the flow of incompressible viscoelastic fluids is established. The primary objective is to formulate the governing equations and constitutive models that describe the physical system within the four-roll mill geometry. We begin from the universal conservation principles of mass and linear momentum, introducing the physical and kinematic simplifications characteristic of polymer solutions under creeping and low-inertia flow conditions. The Oldroyd-B constitutive model is formulated by examining the Upper-Convected Time Derivative (UCTD), detailing its extensional kinematics and the origins of the High-Weissenberg Number Problem (HWNP). Non-linear closures, specifically the Giesekus and Phan-Thien–Tanner (PTT) models, are presented as distinct rheological extensions that account for anisotropic hydrodynamic drag and stress-dependent macromolecular entanglement breakdown while preserving the underlying linear structure. A viscous stress scaling procedure is introduced to define the governing dimensionless groups (Re_{scale} , Re_{phys} , Wi , β). Finally, the four-roll mill benchmark is formalized through its local kinematics, flow classification, geometry, boundary conditions, and the definition of

the forward and inverse problems.

1.1 General Conservation Laws

From a continuum mechanics perspective, the spatial and temporal motion of a fluid continuum is governed by the universal conservation laws of mass and linear momentum [1].

1.1.1 Principle of Mass Conservation

The conservation of mass for a fluid with local density $\rho(\mathbf{x}, t)$ and velocity vector $\mathbf{u}(\mathbf{x}, t)$ is expressed through the continuity equation:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0. \quad (1.1)$$

Equation (1.1) dictates that the instantaneous rate of mass accumulation within any fixed control volume must be strictly balanced by the net convective mass flux through its boundary surface.

1.1.2 Conservation of Linear Momentum

Applying Newton's second law to an infinitesimal fluid parcel yields the momentum equation in Eulerian coordinates:

$$\rho \left(\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} \right) = \nabla \cdot \boldsymbol{\sigma} + \rho \mathbf{g}, \quad (1.2)$$

where $\mathbf{g}$ denotes the acceleration due to external body forces, and $\boldsymbol{\sigma}$ is the symmetric second-order stress tensor. The left-hand side represents the substantial derivative $D\mathbf{u}/Dt$, which combines the local unsteady rate of change $\partial \mathbf{u}/\partial t$ with the convective acceleration term $\mathbf{u} \cdot \nabla \mathbf{u}$ arising from spatial velocity gradients.

To isolate the isotropic pressure field from internal stresses generated by deformation, the total stress tensor is partitioned as:

$$\boldsymbol{\sigma} = -p\mathbf{I} + \mathbf{T}, \quad (1.3)$$

where p is the isotropic pressure field, $\mathbf{I}$ is the second-order identity tensor, and $\mathbf{T}$ is the extra-stress tensor. The extra-stress tensor encapsulates all non-hydrostatic stresses associated with the dissipative and elastic response of the fluid microstructure. Substituting Equation (1.3) into Equation (1.2) yields:

$$\rho \left(\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} \right) = -\nabla p + \nabla \cdot \mathbf{T} + \rho \mathbf{g}. \quad (1.4)$$

1.2 Physical Assumptions and Governing System Reduction

The full three-dimensional, unsteady equations of motion can be reduced by introducing scale-appropriate physical assumptions for polymer solutions within the four-roll mill geometry:

1. **Continuous medium and state of rest:** The fluid is modeled as a continuous medium that is locally isotropic in its thermodynamic state of rest. Under kinematic deformation, flow-induced macromolecular orientation and stretching break this resting isotropy, generating stress anisotropies captured by the extra-stress tensor.
2. **Incompressibility and stream function formulation:** Under the moderate operating pressures and low flow velocities characteristic of liquid polymer systems, density variations are negligible ($\rho = \text{const}$). The mass

conservation equation (1.1) simplifies to the divergence-free condition:

$$\nabla \cdot \mathbf{u} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0. \quad (1.5)$$

In two-dimensional planar motion, this incompressibility constraint can be satisfied identically by introducing a scalar stream function $\psi(x, y)$ defined such that:

$$u = \frac{\partial \psi}{\partial y}, \quad v = -\frac{\partial \psi}{\partial x}. \quad (1.6)$$

By Schwarz's theorem on mixed partial derivatives, $\nabla \cdot \mathbf{u} = \frac{\partial^2 \psi}{\partial x \partial y} - \frac{\partial^2 \psi}{\partial y \partial x} \equiv 0$ holds identically across the entire domain.

3. **Isothermal operating conditions:** Under the operating conditions considered in this study, internal temperature variations due to viscous dissipation remain negligible. The energy conservation equation decouples from momentum, allowing transport properties (viscosities, relaxation time) to be treated as constants.
4. **Absence of body forces:** The four-roll mill geometry is analyzed in a horizontal cross-sectional plane (x, y) . Gravitational acceleration acts along the vertical cylinder axis (z) , perpendicular to the modeled planar flow field. Consequently, body force contributions vanish in the modeled plane, and we set $\mathbf{g} = \mathbf{0}$.
5. **Steady-state flow:** Once constant operating conditions are established and the flow reaches a steady regime, all local temporal derivatives vanish identically ($\partial(\cdot)/\partial t = \mathbf{0}$).
6. **Two-dimensional planar kinematics:** Given the high aspect ratio of the cylindrical rollers along the axial dimension (z -axis), edge effects in the transverse mid-plane of the cavity are negligible. The velocity field is

therefore strictly planar:

$$\mathbf{u} = (u(x, y), v(x, y), 0). \quad (1.7)$$

Under these physical simplifications, the momentum equation reduces to:

$$\rho(\mathbf{u} \cdot \nabla \mathbf{u}) = -\nabla p + \nabla \cdot \mathbf{T}. \quad (1.8)$$

1.3 Constitutive Modeling of Viscoelastic Fluids

In two dimensions, the momentum balance (1.8) contains six unknown scalar fields $(u, v, p, T_{xx}, T_{xy}, T_{yy})$, whereas mass and momentum conservation provide only three equations. A constitutive relationship for the extra-stress tensor $\mathbf{T}$ is required to close the system.

For dilute and semi-dilute polymer solutions, the extra-stress tensor $\mathbf{T}$ is partitioned into a purely viscous Newtonian solvent contribution and an elastic polymeric contribution $\boldsymbol{\tau}$ [1, 2]:

$$\mathbf{T} = 2\mu_s \mathbf{D} + \boldsymbol{\tau}, \quad (1.9)$$

where μ_s represents the dynamic viscosity of the Newtonian solvent, and $\mathbf{D}$ denotes the symmetric rate-of-strain tensor:

$$\mathbf{D} = \frac{1}{2} [\nabla \mathbf{u} + (\nabla \mathbf{u})^T]. \quad (1.10)$$

This decomposition isolates the instantaneous dissipative response of the solvent ($2\mu_s \mathbf{D}$) from the elastic stress $\boldsymbol{\tau}$ generated by macromolecular stretching and

orientation. Substituting Equation (1.9) into Equation (1.8) and exploiting incompressibility ($\nabla \cdot (2\mu_s \mathbf{D}) = \mu_s \nabla^2 \mathbf{u}$), the steady momentum equation becomes:

$$\rho(\mathbf{u} \cdot \nabla \mathbf{u}) = -\nabla p + \mu_s \nabla^2 \mathbf{u} + \nabla \cdot \boldsymbol{\tau}. \quad (1.11)$$

1.3.1 Upper-Convected Time Derivative

Polymeric liquids exhibit memory: macromolecular chains deform, orient, and relax over characteristic finite timescales. When a fluid element moves through a complex flow, its internal stress state is simultaneously advected along streamlines, rotated by local vorticity, and stretched by velocity gradients.

To satisfy the principle of material frame-indifference, constitutive equations must employ a frame-invariant tensor rate. For flexible polymer chains undergoing affine deformation with the surrounding continuum, the appropriate objective derivative is the *Upper-Convected Time Derivative* (UCTD), denoted by $\overset{\nabla}{\boldsymbol{\tau}}$. With the velocity gradient defined as $(\nabla \mathbf{u})_{ij} = \partial u_i / \partial x_j$, the upper-convected derivative of the symmetric extra-stress tensor $\boldsymbol{\tau}$ is:

$$\overset{\nabla}{\boldsymbol{\tau}} \equiv \frac{\partial \boldsymbol{\tau}}{\partial t} + \mathbf{u} \cdot \nabla \boldsymbol{\tau} - (\nabla \mathbf{u}) \boldsymbol{\tau} - \boldsymbol{\tau} (\nabla \mathbf{u})^T. \quad (1.12)$$

Under steady conditions, $\partial \boldsymbol{\tau} / \partial t = \mathbf{0}$. The term $\mathbf{u} \cdot \nabla \boldsymbol{\tau}$ accounts for convective spatial transport of stress along flow streamlines, while the terms $-(\nabla \mathbf{u}) \boldsymbol{\tau} - \boldsymbol{\tau} (\nabla \mathbf{u})^T$ describe the stretching and rotation of macromolecular conformations induced by local velocity gradients.

1.3.2 The Oldroyd-B Model and Extensional Behavior

The classical Oldroyd-B constitutive model [4] represents macromolecules as dilute, non-interacting Hookean dumbbells suspended in a Newtonian solvent [1].

In dilute solution theory, the polymeric viscosity contribution μ_p is directly coupled to the relaxation time λ through kinetic theory:

$$\mu_p = nk_B T \lambda, \quad (1.13)$$

where n is the number density of polymer chains, k_B is the Boltzmann constant, and T is absolute temperature. Under steady conditions, the polymeric extra-stress tensor $\boldsymbol{\tau}$ satisfies:

$$\boldsymbol{\tau} + \lambda \overset{\nabla}{\boldsymbol{\tau}} = 2\mu_p \mathbf{D}. \quad (1.14)$$

The total zero-shear viscosity of the fluid is $\mu_0 = \mu_s + \mu_p$.

While Oldroyd-B predicts a constant shear viscosity $\eta(\dot{\gamma}) = \mu_0$ (characteristic of ideal elastic fluids), it exhibits an intrinsic limitation in extensional kinematics. In a steady planar extensional flow ($u = Ex, v = -Ey$, with extension rate $E > 0$), as the extension rate approaches the critical threshold:

$$E_{\text{crit}} = \frac{1}{2\lambda}, \quad (1.15)$$

the assumption of infinitely extendable Hookean dumbbells allows unbounded stretching. Consequently, the polymeric stress and extensional viscosity diverge asymptotically to infinity ($\eta_E \rightarrow \infty$). At low deformation rates ($E \rightarrow 0$), the planar extensional viscosity recovers the Newtonian limit, yielding a planar Trouton ratio $\text{Tr}_E = \eta_E/\mu_0 \rightarrow 4$.

This unphysical stress divergence near the critical threshold $2\lambda E \rightarrow 1$ is the physical manifestation of the *High-Weissenberg Number Problem* (HWNP), which causes severe numerical breakdown in conventional grid-based solvers due to exponentially steep stress boundary layers [3].

1.3.3 Non-linear Closures: Giesekus and Phan-Thien–Tanner (PTT)

To overcome the unphysical stress singularity of Oldroyd-B and capture the behavior of semi-dilute solutions and concentrated polymer melts, non-linear constitutive models introduce stress-limiting mechanisms. The Giesekus and Phan-Thien–Tanner (PTT) models represent two distinct extensions of the baseline Oldroyd-B framework, incorporating different physical mechanisms to modify the rheological response at high deformation rates while retaining the same linear baseline at low strain rates:

The Giesekus Model

The Giesekus constitutive equation [1,20] accounts for anisotropic Brownian drag and hydrodynamic interactions between neighboring macromolecules in concentrated systems. This is achieved by introducing a quadratic stress relaxation term:

$$\boldsymbol{\tau} + \lambda \overset{\nabla}{\boldsymbol{\tau}} + \frac{\alpha \lambda}{\mu_p} \boldsymbol{\tau}^2 = 2\mu_p \mathbf{D}, \quad (1.16)$$

where $\alpha \in [0, 0.5]$ is the dimensionless mobility parameter, and the tensorial square is defined by matrix contraction $(\boldsymbol{\tau}^2)_{ij} = \tau_{ik}\tau_{kj}$. When $\alpha = 0$, the model reduces to Oldroyd-B. For $\alpha > 0$, the quadratic term limits the growth of extensional stresses, preventing the unphysical divergence of Oldroyd-B and introducing shear-thinning behavior in steady shear flows.

The Phan-Thien–Tanner (PTT) Model

Derived from the transient network theory of concentrated polymeric systems [1,21], the PTT model assumes that polymer chains form temporary physical entanglements whose rate of creation and destruction depends on the stored elastic

stress. In the linear PTT formulation, the constitutive equation is:

$$f(\boldsymbol{\tau})\boldsymbol{\tau} + \lambda \overset{\nabla}{\boldsymbol{\tau}} = 2\mu_p \mathbf{D}, \quad (1.17)$$

where the stress-softening function $f(\boldsymbol{\tau})$ depends linearly on the trace of the extra-stress tensor:

$$f(\boldsymbol{\tau}) = 1 + \frac{\varepsilon\lambda}{\mu_p} \text{tr}(\boldsymbol{\tau}), \quad (1.18)$$

with $\text{tr}(\boldsymbol{\tau}) = \tau_{xx} + \tau_{yy}$, and $\varepsilon \geq 0$ representing the dimensionless extensibility parameter controlling the stress-induced rate of entanglement breakdown. When $\varepsilon = 0$, $f = 1$ and the model recovers Oldroyd-B. For $\varepsilon > 0$, the effective relaxation rate increases with stress, bounding the extensional viscosity.

Unified Framework for Inverse Identification

These constitutive closures share the common baseline structure of Oldroyd-B while differing through distinct non-linear mechanisms parameterized by α and ε . This common mathematical structure is particularly advantageous for inverse parameter identification, as the model can evaluate whether the response of a fluid flow aligns with quasi-linear dumbbell elasticity ($\alpha = 0, \varepsilon = 0$), anisotropic drag ($\alpha > 0$), or network breakdown ($\varepsilon > 0$).

1.4 Nondimensionalization and Governing Equations

Nondimensionalization scales physical variables to order unity and establishes the governing dimensionless groups [3]. In low-Reynolds viscoelastic flows, scaling by a viscous stress unit rather than an inertial dynamic pressure improves numerical conditioning and balances the magnitudes of the governing differential operators.

1.4.1 Reference Scales and Dimensionless Groups

We select reference length H_{ref} (roller radius R), reference velocity U_{ref} (roller surface speed u_{roll}), and a constant reference viscosity η_0 (chosen as a fixed baseline scale, e.g., $\eta_0 = 1.0 \text{ Pa} \cdot \text{s}$). The dimensionless variables are defined as:

$$\mathbf{x}^* = \frac{\mathbf{x}}{H_{\text{ref}}}, \quad \mathbf{u}^* = \frac{\mathbf{u}}{U_{\text{ref}}}, \quad p^* = \frac{p}{\frac{\eta_0 U_{\text{ref}}}{H_{\text{ref}}}}, \quad \boldsymbol{\tau}^* = \frac{\boldsymbol{\tau}}{\frac{\eta_0 U_{\text{ref}}}{H_{\text{ref}}}}. \quad (1.19)$$

Dropping asterisks for notational brevity, the governing non-dimensional groups are:

$$Re_{\text{scale}} = \frac{\rho U_{\text{ref}} H_{\text{ref}}}{\eta_0} \quad (\text{Scaling Reynolds number}), \quad (1.20)$$

$$Re_{\text{phys}} = \frac{\rho U_{\text{ref}} H_{\text{ref}}}{\mu_s + \mu_p} = \frac{Re_{\text{scale}}}{\mu_s^* + \mu_p^*} \quad (\text{Physical Reynolds number}), \quad (1.21)$$

$$Wi = \frac{\lambda U_{\text{ref}}}{H_{\text{ref}}} \quad (\text{Global Weissenberg number}), \quad (1.22)$$

$$\beta = \frac{\mu_s}{\mu_s + \mu_p} = \frac{\mu_s^*}{\mu_s^* + \mu_p^*} \quad (\text{Viscosity ratio}), \quad (1.23)$$

$$\mu_s^* = \frac{\mu_s}{\eta_0}, \quad \mu_p^* = \frac{\mu_p}{\eta_0} \quad (\text{Dimensionless solvent and polymer viscosities}). \quad (1.24)$$

The distinction between Re_{scale} and Re_{phys} is essential in inverse problems: defining the momentum equation through the constant scale Re_{scale} ensures that the numerical scaling of convective acceleration remains invariant even as unknown fluid viscosities (μ_s, μ_p) are adjusted during optimization.

1.4.2 Dimensionless Vector Formulation

The non-dimensional governing equations in steady 2D flow become:

$$\nabla \cdot \mathbf{u} = 0, \quad (1.25)$$

$$Re_{\text{scale}}(\mathbf{u} \cdot \nabla \mathbf{u}) + \nabla p - \mu_s^* \nabla^2 \mathbf{u} - \nabla \cdot \boldsymbol{\tau} = \mathbf{0}, \quad (1.26)$$

$$\mathcal{G}(\boldsymbol{\tau}) + Wi (\mathbf{u} \cdot \nabla \boldsymbol{\tau} - (\nabla \mathbf{u}) \boldsymbol{\tau} - \boldsymbol{\tau} (\nabla \mathbf{u})^T) - 2\mu_p^* \mathbf{D} = \mathbf{0}, \quad (1.27)$$

where the constitutive operator $\mathcal{G}(\boldsymbol{\tau})$ unifies the models under study:

$$\mathcal{G}(\boldsymbol{\tau}) = \begin{cases} \boldsymbol{\tau} & \text{(Oldroyd-B),} \\ \boldsymbol{\tau} + \frac{\alpha Wi}{\mu_p^*} \boldsymbol{\tau}^2 & \text{(Giesekus),} \\ \left(1 + \frac{\varepsilon Wi}{\mu_p^*} \text{tr}(\boldsymbol{\tau})\right) \boldsymbol{\tau} & \text{(Linear PTT).} \end{cases} \quad (1.28)$$

1.5 The Four-Roll Mill Benchmark Kinematics and Boundaries

The four-roll mill is a canonical benchmark designed to generate controlled two-dimensional planar extensional flow with a central stagnation point [5–7, 17]. Originally conceived by Taylor [5] and systematically developed by Fuller and Leal [7, 17], it has served as a standard apparatus for analyzing macromolecular deformation and flow-induced birefringence. As depicted in Figure 1.1, the setup consists of four cylindrical rollers placed symmetrically inside a square domain.

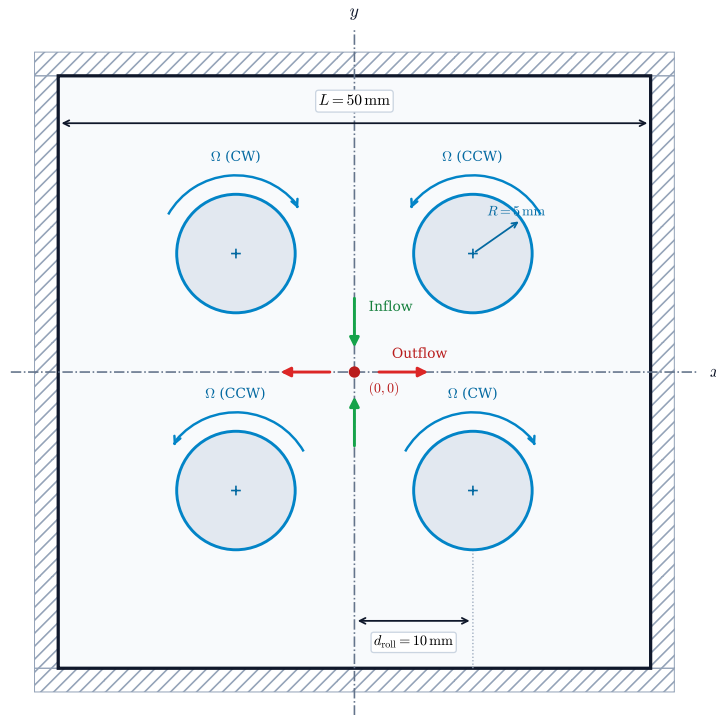


Figure 1.1: Schematic representation of the four-roll mill geometry, illustrating the square cavity of side L , cylindrical rollers of radius R at offset d_{roll} , Cartesian axes (x, y) , rotation directions with angular velocity Ω (adjacent rolls counter-rotate, diagonal rolls co-rotate), and the central hyperbolic stagnation point $(0, 0)$.

1.5.1 Kinematic Relations and Flow Classification

In the vicinity of the central region between the rollers, the local velocity field can be represented by a linear superposition of extension and rotation [7, 17]:

$$\mathbf{u}(x, y) = \begin{pmatrix} Ex + \omega y \\ -Ey - \omega x \end{pmatrix}, \quad (1.29)$$

where $E \geq 0$ denotes the extension rate and $\omega \geq 0$ represents the local angular rotation rate. The associated scalar vorticity of this field is $\omega_z = (\nabla \times \mathbf{u})_z = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} = -2\omega$. To quantify the relative magnitude of extension versus rotation, a dimensionless flow type parameter $\chi \in [-1, 1]$ is defined:

$$\chi = \frac{E - \omega}{E + \omega}. \quad (1.30)$$

Depending on the roller speeds and rotation directions, this idealized local kinematics captures three canonical flow regimes (illustrated in Figure 1.2):

- $\chi = 1$ ($\omega = 0$): **Pure Planar Extensional Flow.** Generated when adjacent rolls counter-rotate symmetrically (diagonal rolls co-rotate). Fluid elements experience pure stretching along x and pure compression along y , with hyperbolic streamlines $xy = \text{const.}$
- $\chi = 0$ ($E = \omega$): **Simple Shear Flow.** An intermediate state where rate of strain and angular rotation have equal magnitude ($E = \omega$), producing rectilinear shear streamlines. In an enclosed cavity, this uniform shear field is realized locally near the center, while further away the flow curves to adapt to roller and wall boundaries.
- $\chi = -1$ ($E = 0$): **Pure Solid-Body Rotation.** Generated when rolls co-rotate to induce bulk circulation, yielding concentric circular streamlines

around the origin representing rigid-body rotation without elongation or compression.

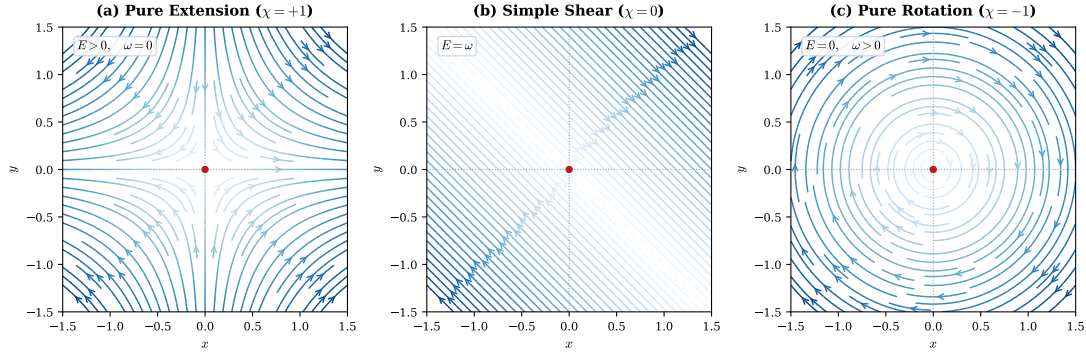


Figure 1.2: Streamline patterns of the two-dimensional linear flow field in Equation (1.29) for the three canonical regimes of the flow classification parameter χ : (a) pure planar extension ($\chi = +1$), (b) simple shear ($\chi = 0$), and (c) pure solid-body rotation ($\chi = -1$). The red marker indicates the central stagnation point $(0, 0)$.

1.5.2 The Central Hyperbolic Stagnation Point

In the extensional regime ($\chi = 1$), the velocity vanishes at the geometric center:

$$\mathbf{u}(0, 0) = \mathbf{0}. \quad (1.31)$$

The velocity gradient tensor at the origin,

$$\nabla \mathbf{u} = \begin{pmatrix} E & 0 \\ 0 & -E \end{pmatrix}, \quad (1.32)$$

possesses real eigenvalues $+E$ (along the horizontal stretching axis x) and $-E$ (along the vertical compression axis y), defining a hyperbolic saddle-point geometry. Along the incoming vertical streamline ($x = 0$), fluid parcels decelerate toward zero velocity, providing fluid elements with long residence times.

This kinematic feature allows macromolecular deformation to accumulate over timescales comparable to or larger than the fluid relaxation time λ , providing an ideal testbed for observing elastic stress growth away from solid boundaries.

1.5.3 Geometric Setup and Operating Parameters

Following the established benchmark setup of D’Avino et al. [18], the physical configuration is characterized by:

- **Cavity dimensions:** Square enclosure of side $L = 50$ mm spanning $[-25, 25]$ mm;
- **Cylindrical rollers:** Radius $R = 5$ mm ($H_{\text{ref}} = R = 0.1L$);
- **Roller placement:** Centers positioned symmetrically at $(\pm 10, \pm 10)$ mm ($d_{\text{roll}} = 10$ mm);
- **Roller kinematics:** Peripheral surface velocity $u_{\text{roll}} = \Omega R = 8.333 \times 10^{-3}$ m/s;
- **Non-dimensional domain:** In units of $H_{\text{ref}} = R$, the computational domain spans $[-5, 5]^2$, with roller centers located at $(\pm 2.0, \pm 2.0)$ and non-dimensional roller radius $R^* = 1.0$.

The physical and operating parameters investigated in this work are summarized in Table 1.1.

1.5.4 Boundary Conditions and Pressure Point Anchoring

To complete the continuous boundary value formulation, physical boundary conditions must be specified along all solid surfaces:

1. **Tangential Velocity on Roller Walls (Γ_{rolls}):** On the surface of each roller $k \in \{\text{TL}, \text{TR}, \text{BL}, \text{BR}\}$, the fluid velocity satisfies no-slip with the

Table 1.1: Physical and operating parameters for the four-roll mill benchmark.

Parameter	Symbol	Value / Range	Unit
Cavity side length	L	0.05	m
Roller radius (reference length)	$H_{\text{ref}} = R$	0.005	m
Roller surface velocity	u_{roll}	8.333×10^{-3}	m/s
Fluid density	ρ	1000	kg/m ³
Characteristic viscosity scale	η_0	1.0	Pa · s
Polymer relaxation time	λ	0.05–1.0	s
Viscosity ratio	β	0.1–0.5	—
Scaling Reynolds number	Re_{scale}	$0.0417 \ll 1$	—
Weissenberg number	Wi	0.083–1.667	—

prescribed peripheral rotation:

$$\mathbf{u}|_{\Gamma_k} = \boldsymbol{\Omega}_k \times (\mathbf{x} - \mathbf{x}_k), \quad (1.33)$$

where $\mathbf{x}_k$ is the roller center. For the planar extensional regime ($\chi = 1$), adjacent rolls counter-rotate while diagonal rolls co-rotate: top-left and bottom-right rolls rotate clockwise ($-\Omega$), while top-right and bottom-left rolls rotate counter-clockwise ($+\Omega$).

2. **No-Slip Velocity on Outer Cavity Walls** (Γ_{outer}): Along the stationary external boundaries of the square enclosure, the velocity vanishes:

$$\mathbf{u}|_{\Gamma_{\text{outer}}} = \mathbf{0}. \quad (1.34)$$

3. **Pressure Point Anchoring**: Because pressure appears in the momentum balance (1.26) exclusively through its gradient ∇p , in an enclosed cavity with prescribed velocity Dirichlet boundary conditions the pressure is determined only up to an arbitrary additive constant ($p \rightarrow p + C$). To fix this gauge freedom and obtain a unique solution, a single reference pressure point is specified:

$$p(\mathbf{x}_0) = 0, \quad (1.35)$$

at an arbitrary reference point $\mathbf{x}_0$ within the domain.

1.6 Problem Formulation and Computational Transition

The physical laws, constitutive closures, and boundary constraints established in this chapter define two complementary problem settings:

1.6.1 Forward and Inverse Problem Formulations

- **Forward (Direct) Problem:** Given a complete set of known fluid parameters $(\lambda, \mu_p, \mu_s, \alpha, \varepsilon)$ and the prescribed kinematic boundary conditions on rolls and walls, determine the velocity field $\mathbf{u}(x, y)$, pressure field $p(x, y)$, and polymeric extra-stress tensor $\boldsymbol{\tau}(x, y)$ that satisfy the momentum balance (1.26) and constitutive closure (1.27) throughout the cavity domain Ω .
- **Inverse (Parameter Identification) Problem:** Given partial kinematic observations of the velocity field within the cavity, reconstruct the latent internal fields $(p, \boldsymbol{\tau})$ and identify the unknown physical parameters governing the fluid $(\lambda, \mu_p, \mu_s, \text{alongside non-linear parameters } \alpha \text{ or } \varepsilon)$.

Table 1.2 classifies the role of each variable and parameter across the two problem formulations.

1.6.2 Transition to Physics-Informed Neural Networks

The differential system formulated in Equations (1.25)–(1.28) represents a highly coupled, non-linear system of partial differential equations. In Chapter 2, this continuous physical formulation is translated into a machine-learning framework

Table 1.2: Classification of physical fields and parameters across forward and inverse problem formulations.

Physical Quantity	Symbol	Forward Problem	Inverse Problem
Velocity Field	$\mathbf{u} = (u, v)$	Primary unknown (via ψ)	Observed / Superficial
Pressure Field	p	Primary unknown	Latent (unobserved)
Polymeric Stress	$\boldsymbol{\tau} = (\tau_{xx}, \tau_{xy}, \tau_{yy})$	Primary unknown	Latent (unobserved)
Relaxation Time	λ (Wi)	Known parameter	Unknown physical parameter
Polymeric Viscosity	μ_p	Known parameter	Unknown physical parameter
Solvent Viscosity	μ_s	Known parameter	Unknown physical parameter
Giesekus Mobility	α	Known parameter	Unknown parameter / Model choice
PTT Extensibility	ε	Known parameter	Unknown parameter / Model choice

using Physics-Informed Neural Networks (PINNs). By embedding the momentum and constitutive equations directly into the neural network loss function through automatic differentiation, the PINN framework enables both forward flow reconstruction and inverse rheological parameter discovery without requiring traditional spatial grids.

Chapter 2

Physics-Informed Neural Networks in Complex Rheology

Feedforward deep networks, commonly known as Multi-Layer Perceptrons (MLPs), represent the quintessential architecture of deep learning [15]. Conceptually, an MLP is a parametric function defined by composing together a sequence of simpler parametric functions. Each of these constituent functions has multiple inputs and outputs, and in neural network terminology is called a *layer*. The scalar outputs computed within each layer are referred to as *units* or *features*. Although each unit performs a relatively simple mathematical mapping of its inputs, chaining them in succession allows the network as a whole to express continuous functions of arbitrary complexity.

The structural dimensions of a network are conventionally characterized by its *width*—the number of units in a given layer—and its *depth*—the total number of layers. Having significant width and depth provides deep architectures with remarkable representational power, making them the foundational technology behind modern machine learning applications.

To appreciate the paradigm shift introduced by deep neural networks, it is useful to contrast them with traditional machine learning algorithms, such as

linear regression, logistic classifiers, and kernel methods. Classical algorithms operate by applying a linear transformation to a fixed set of features. These methods can indeed learn non-linear behaviors, but their non-linearity is rigid: it is non-linear only in the original input space, while remaining strictly linear in a feature space that must be pre-defined and engineered manually by human experts. Neural networks fundamentally overcome this limitation by automating the design of features. Rather than relying on manual feature engineering, an MLP learns the optimal representations directly from data.

In this architecture, the layers between the inputs and the final outputs are termed *hidden layers*. They are called hidden because their internal values are not specified or observed in the training data; instead, the optimization algorithm must autonomously determine how to organize these intermediate representations to solve the target problem. By transforming inputs through successive hidden layers, neural networks can approximate complex continuous relationships.

In this chapter, we explore how these feedforward architectures form the computational core of scientific machine learning. We begin by reviewing the fundamentals of neural networks and their optimization. We then examine why continuous neural representations are compelling for physical modeling, while discussing the critical limitations of purely data-driven black-box approaches. This analysis naturally motivates the Physics-Informed Neural Network (PINN) paradigm, followed by a review of foundational literature benchmarks and the specific research objectives of this dissertation in complex viscoelastic flows.

2.1 Fundamentals of Deep Learning

Deep learning algorithms approximate continuous mathematical functions through multi-layer architectures that learn internal representations from data [15]. In this section, we present the essential building blocks of these networks, focus-

ing on their mathematical structure, smooth activation functions, optimization algorithms, and gradient evaluation.

2.1.1 The Artificial Neuron and Non-Linear Activation Functions

The elementary processing unit of a neural network is the artificial neuron. Given an input vector $\mathbf{x}$, a neuron computes an affine combination of the inputs followed by a scalar non-linear mapping:

$$z = \mathbf{w} \cdot \mathbf{x} + b, \quad a = \sigma(z), \quad (2.1)$$

where $\mathbf{w}$ is the weight vector, b is a scalar bias, and $\sigma(\cdot)$ is the activation function. The activation function introduces non-linearity into the model. Without non-linear activations, composing multiple linear layers would simply collapse mathematically into another single linear mapping, eliminating the expressive advantage of deep architectures.

In standard deep learning applications such as image recognition, piecewise-linear activations—most notably the Rectified Linear Unit ($\text{ReLU}(z) = \max(0, z)$)—are widely adopted due to their computational simplicity. However, in scientific computing and fluid mechanics, the network’s outputs represent continuous physical fields governed by partial differential equations (PDEs), such as velocities, pressures, and stresses. These governing laws involve spatial derivatives of first, second, or higher order (such as viscous laplacians $\nabla^2 \mathbf{u}$ or stress divergences $\nabla \cdot \boldsymbol{\tau}$).

Because the first derivative of ReLU is discontinuous and its second derivative is zero almost everywhere ($\text{ReLU}''(z) \equiv 0$), evaluating a ReLU network inside differential equations would yield zero for diffusive and curvature terms. For this reason, physics-informed models strictly require smooth, infinitely differentiable

activation functions ($\mathcal{C}^\infty$). In this work, we employ the Sigmoid Linear Unit (SiLU, also known as Swish) [15]:

$$\text{SiLU}(z) = \frac{z}{1 + e^{-z}}. \quad (2.2)$$

The SiLU activation is smooth, non-monotonic, and self-gated. Unlike classical functions such as the hyperbolic tangent ($\tanh$) that saturate toward constant limits for large inputs, SiLU remains active in the positive regime. This property helps maintain healthy gradient flow across regions characterized by intense localized shear and steep stress variations, which are typical of viscoelastic flows.

2.1.2 Multi-Layer Perceptron Architecture

By arranging artificial neurons into sequential layers, we construct a Multi-Layer Perceptron (MLP). The network maps an input vector $\mathbf{x}$ to an output prediction $\hat{\mathbf{u}}$ by propagating information forward through L successive layers:

$$\begin{aligned} \mathbf{h}^{(1)} &= \sigma \left(\mathbf{W}^{(1)} \mathbf{x} + \mathbf{b}^{(1)} \right), \\ \mathbf{h}^{(l)} &= \sigma \left(\mathbf{W}^{(l)} \mathbf{h}^{(l-1)} + \mathbf{b}^{(l)} \right), \quad \text{for } l = 2, \dots, L-1, \\ \hat{\mathbf{u}} &= \mathbf{W}^{(L)} \mathbf{h}^{(L-1)} + \mathbf{b}^{(L)}, \end{aligned} \quad (2.3)$$

where $\mathbf{h}^{(l)}$ denotes the hidden feature vector at layer l , and $\mathbf{W}^{(l)}$ and $\mathbf{b}^{(l)}$ represent the weight matrix and bias vector of that layer. In the context of this thesis, the inputs $\mathbf{x} = (x, y)$ are spatial coordinates within the fluid domain, while the outputs $\hat{\mathbf{u}}$ correspond to the continuous physical fields of interest, such as the stream function ψ , the pressure p , and the extra-stress tensor components $\boldsymbol{\tau}$.

A central theoretical property of MLPs is their universal approximation capability [14, 23]: a feedforward network with sufficient width and smooth non-linear activations can approximate any continuous function on a closed domain to an

arbitrary degree of accuracy.

Weight Initialization: Glorot Scheme and Output Zero-Initialization

The initialization of the network weights and biases strongly influences optimization stability. To avoid vanishing or exploding signals across deep layers, the hidden layer weights are initialized using the Xavier/Glorot normal distribution [15], which scales the variance of the initial parameters according to the width of adjacent layers.

For the final output layers representing the extra-stress tensor $\boldsymbol{\tau}$ and the pressure p , we adopt a strategic zero-initialization ($\mathbf{W}^{(L)} = \mathbf{0}, \mathbf{b}^{(L)} = \mathbf{0}$). This choice is motivated by two practical reasons:

1. **Thermodynamic Baseline:** In fluid mechanics, a fluid at thermodynamic rest has zero extra-stress ($\boldsymbol{\tau} = \mathbf{0}$). Zero-initialization guarantees that at iteration zero, the network outputs correspond to the unperturbed resting state.
2. **Suppression of Numerical Shocks:** If output weights were randomly initialized, initial stress predictions would exhibit high-frequency spatial noise across the domain. Evaluating these random fluctuations inside non-linear constitutive equations—which contain quadratic stress products and convective gradients—would cause initial residual losses to spike violently, destabilizing the optimization. Initializing the output layer to zero ensures smooth, bounded initial residuals.

2.1.3 Loss Functions and Optimization Algorithms

Training a neural network consists of finding the optimal set of parameters $\boldsymbol{\theta} = \{\mathbf{W}^{(l)}, \mathbf{b}^{(l)}\}$ that minimizes a scalar loss function $\mathcal{L}(\boldsymbol{\theta})$. The loss measures how

well the network's predictions match desired targets, typically using the Mean Squared Error (MSE):

$$\mathcal{L}_{\text{MSE}}(\boldsymbol{\theta}) = \frac{1}{N} \sum_{i=1}^N \|\hat{\mathbf{u}}(\mathbf{x}_i; \boldsymbol{\theta}) - \mathbf{u}_i^{\text{target}}\|_2^2. \quad (2.4)$$

The parameters are updated iteratively in the direction that reduces the loss, following the principle of gradient descent:

$$\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \eta \nabla_{\boldsymbol{\theta}} \mathcal{L}, \quad (2.5)$$

where $\eta > 0$ is the learning rate, which controls the step size taken along the negative gradient.

In this work, training is carried out using a two-stage optimization strategy:

1. **First-Order Optimization (Adam):** The Adaptive Moment Estimation optimizer [22] dynamically adjusts the learning rate for each individual parameter based on running estimates of past gradients. This makes Adam robust and effective at exploring complex loss landscapes during the early stages of training.
2. **Second-Order Quasi-Newton Optimization (L-BFGS):** Once the solution is close to a minimum, the Limited-memory Broyden–Fletcher–Goldfarb–Shanno (L-BFGS) algorithm uses curvature information to achieve faster, super-linear convergence. In scientific problems, L-BFGS is crucial for refining the solution and driving physical residuals down to high accuracy ($10^{-5} - 10^{-7}$).

2.1.4 Backpropagation and Computational Graphs

To update the parameters via gradient descent, the network must compute the derivative of the loss with respect to every weight and bias, denoted by $\nabla_{\theta}\mathcal{L}$. Evaluating these derivatives numerically by perturbing each parameter one by one would be computationally infeasible.

Instead, neural networks compute exact gradients efficiently using the *backpropagation* algorithm [15]. The execution of a neural network can be viewed as a directed computational graph, where inputs flow forward through mathematical operations to produce the predictions and the final loss value. Backpropagation applies the classical chain rule of calculus in reverse order, starting from the loss and propagating sensitivities backward through the graph.

Through this reverse-mode evaluation, the gradients with respect to all parameters in the network are computed simultaneously in a single backward pass, at a computational cost roughly equivalent to that of a single forward evaluation. This computational efficiency is what makes training deep models practical.

2.2 Why Neural Networks? Capabilities and Physical Limitations

2.2.1 Capabilities and Universal Function Approximation

Neural networks offer distinctive characteristics that make them attractive for continuous physical modeling:

1. **Continuous and Meshless Representation:** Traditional grid-based methods (Finite Element, Finite Volume, and Finite Difference Methods) represent state variables exclusively at discrete vertices or cell centers, relying on explicit mesh connectivity. In contrast, a neural network provides a contin-

uous, globally defined mapping $\mathbf{x} \mapsto \hat{\mathbf{u}}(\mathbf{x})$ over the entire domain, capable of being evaluated at arbitrary spatial coordinates. However, being mesh-less does not imply freedom from geometric considerations: defining the domain, enforcing boundary conditions, and designing effective collocation point distributions remain essential and influential steps.

2. **High-Dimensional Functional Approximation:** The universal approximation capacity allows neural networks to represent complex, coupled physical fields (velocities, pressures, and stress tensors) within a unified functional representation, without requiring specialized interpolation elements for each variable.
3. **Hardware Acceleration and Differentiability:** Modern deep learning frameworks are built upon dense linear algebra operations optimized for parallel accelerator hardware (GPUs and TPUs), and provide built-in automatic differentiation for computing spatial derivatives of the represented fields.

2.2.2 Critical Bottlenecks of Purely Data-Driven Machine Learning

Despite their high expressivity, deploying purely data-driven, black-box neural networks to fluid mechanics and non-Newtonian rheology reveals fundamental limitations:

- **Severe Data Scarcity in Experimental Rheology:** Deep neural networks conventionally rely on extensive labeled datasets. In experimental fluid mechanics, acquiring dense velocity fields via techniques such as Particle Image Velocimetry (PIV) is labor-intensive, while non-intrusive measurements of internal extra-stress components are generally unavailable in

complex geometries.

- **Violation of Fundamental Conservation Laws:** Purely data-driven models minimize statistical discrepancy over discrete training points without enforcing physical principles. Consequently, their predictions routinely violate mass conservation ($\nabla \cdot \mathbf{u} \neq 0$), linear momentum balance, and constitutive relationships, leading to unphysical artifacts—particularly when interpolating across sparse regions or extrapolating beyond observed conditions.
- **Overfitting and Spurious Oscillations:** In low-data regimes, over-parameterized networks can readily fit experimental noise or sparse sensor readings with minimal training error, while developing high-frequency, non-physical oscillations in unmonitored subdomains.
- **Inability to Identify Governing Parameters:** Black-box architectures fit mapping functions between inputs and outputs, but cannot infer underlying physical material properties—such as fluid relaxation times or polymeric viscosities—that govern the flow dynamics.

These fundamental shortcomings emphasize that data alone are insufficient for scientific modeling in rheology, motivating frameworks that embed the governing physical equations directly into the learning process.

2.3 The Physics-Informed Neural Network Paradigm

The Physics-Informed Neural Network (PINN) framework, formalized by Raissi, Perdikaris, and Karniadakis [8], addresses the limitations of purely data-driven modeling by embedding mathematical governing equations directly into the objective function of a neural network.

2.3.1 Physics-Constrained Loss Formulation

Consider a general non-linear differential operator $\mathcal{F}$ governing a continuous physical state $\mathbf{u}(\mathbf{x})$ over a spatial domain $\Omega \subset \mathbb{R}^d$ bounded by $\partial\Omega$:

$$\mathcal{F}[\mathbf{u}(\mathbf{x}); \boldsymbol{\lambda}] = \mathbf{0}, \quad \mathbf{x} \in \Omega, \quad (2.6)$$

$$\mathcal{B}[\mathbf{u}(\mathbf{x})] = \mathbf{g}(\mathbf{x}), \quad \mathbf{x} \in \partial\Omega, \quad (2.7)$$

where $\boldsymbol{\lambda}$ represents physical parameters (such as fluid viscosities or relaxation times) and $\mathcal{B}$ denotes boundary differential operators.

In a PINN, the unknown solution $\mathbf{u}(\mathbf{x})$ is approximated by a neural network $\hat{\mathbf{u}}(\mathbf{x}; \boldsymbol{\theta})$. Substituting $\hat{\mathbf{u}}$ into the governing equations defines the differential residual:

$$\mathbf{f}(\mathbf{x}; \boldsymbol{\theta}, \boldsymbol{\lambda}) \equiv \mathcal{F}[\hat{\mathbf{u}}(\mathbf{x}; \boldsymbol{\theta}); \boldsymbol{\lambda}]. \quad (2.8)$$

The network parameters $\boldsymbol{\theta}$ —and, in inverse problems, the unknown physical parameters $\boldsymbol{\lambda}$ —are optimized by minimizing a composite multi-objective loss function:

$$\mathcal{L}_{\text{total}}(\boldsymbol{\theta}, \boldsymbol{\lambda}) = w_{\text{data}}\mathcal{L}_{\text{data}} + w_{\text{bc}}\mathcal{L}_{\text{bc}} + w_{\text{pde}}\mathcal{L}_{\text{pde}}, \quad (2.9)$$

where:

$$\mathcal{L}_{\text{data}} = \frac{1}{N_{\text{data}}} \sum_{i=1}^{N_{\text{data}}} \left\| \hat{\mathbf{u}}(\mathbf{x}_i^{\text{data}}) - \mathbf{u}_i^{\text{data}} \right\|_2^2, \quad (2.10)$$

$$\mathcal{L}_{\text{bc}} = \frac{1}{N_{\text{bc}}} \sum_{j=1}^{N_{\text{bc}}} \left\| \mathcal{B}[\hat{\mathbf{u}}(\mathbf{x}_j^{\text{bc}})] - \mathbf{g}(\mathbf{x}_j^{\text{bc}}) \right\|_2^2, \quad (2.11)$$

$$\mathcal{L}_{\text{pde}} = \frac{1}{N_{\text{pde}}} \sum_{k=1}^{N_{\text{pde}}} \left\| \mathbf{f}(\mathbf{x}_k^{\text{pde}}; \boldsymbol{\theta}, \boldsymbol{\lambda}) \right\|_2^2. \quad (2.12)$$

Here, $\{\mathbf{x}_i^{\text{data}}\}$ denotes available experimental or numerical measurements, $\{\mathbf{x}_j^{\text{bc}}\}$ represents points sampled along physical boundaries $\partial\Omega$, and $\{\mathbf{x}_k^{\text{pde}}\}$ designates

interior *collocation points* scattered across Ω .

This formulation naturally accommodates two distinct problem settings:

- **Forward Problems:** The physical parameters $\boldsymbol{\lambda}$ are fully known. The network optimizes its weights $\boldsymbol{\theta}$ to solve for the unknown fields $\hat{\mathbf{u}}(\mathbf{x})$, acting as a continuous PDE solver.
- **Inverse Problems:** One or more physical parameters in $\boldsymbol{\lambda}$ are unknown. The network simultaneously reconstructs the physical fields and identifies the unknown parameters by leveraging both available observations $\mathcal{L}_{\text{data}}$ and the physical consistency constraints $\mathcal{L}_{\text{pde}}$.

A critical distinction must be drawn between reconstructing physical fields and identifying governing parameters. A neural network may achieve low training residuals and appear to fit observed velocity fields accurately, yet the inferred parameters may remain inaccurate or poorly identified due to parameter correlations, insufficient kinematic variation in the data, or ill-conditioned loss landscapes. Integrating governing equations provides necessary structure, but does not automatically guarantee parameter identifiability or solution uniqueness.

2.3.2 Deep Dive: Standard Supervised Training vs. PINN Optimization

The operational differences between conventional supervised learning and physics-informed optimization are characterized by several key aspects:

1. **Physical Regularization and Inductive Bias:** In standard supervised training, the network is unconstrained in regions devoid of training data, which allows over-parameterized models to fit noisy samples while producing spurious oscillations. In PINNs, evaluating the differential residual $\mathcal{L}_{\text{pde}}$

across interior collocation points imposes an analytical inductive bias: functions exhibiting non-physical gradients or curvature incur substantial penalties from the governing differential operators. This physical regularization significantly constrains the admissible solution space. However, it should not be viewed as an absolute guarantee against overfitting: if measurement data are noisy or if loss weights $w_{\text{data}}, w_{\text{pde}}$ are poorly balanced, the optimization can still overfit local measurements or converge to unphysical local minima.

2. **Coordinate Automatic Differentiation (AD):** In conventional deep learning, backpropagation employs AD to differentiate the scalar loss with respect to network parameters:

$$\text{Backpropagation (Classical DL): } \frac{\partial \mathcal{L}}{\partial \boldsymbol{\theta}}. \quad (2.13)$$

In PINNs, the exact same reverse-mode automatic differentiation engine is applied to compute spatial derivatives of the predicted state fields with respect to input coordinates:

$$\text{Coordinate AD (PINNs): } \frac{\partial \hat{\mathbf{u}}}{\partial \mathbf{x}}, \quad \frac{\partial^2 \hat{\mathbf{u}}}{\partial \mathbf{x}^2}. \quad (2.14)$$

As summarized in Table 2.1, coordinate AD evaluates spatial derivatives analytically along the computational graph, avoiding the truncation error $\mathcal{O}(\Delta x^2)$ inherent in finite-difference stencils. Nonetheless, coordinate AD remains subject to machine floating-point roundoff, network functional approximation errors, and potential numerical ill-conditioning in regions of steep gradient transitions.

3. **Meshless Nature and Point Sampling:** Unlike traditional CFD tech-

niques that rely on conforming discretization meshes and element connectivity tables, PINNs evaluate differential residuals at unorganized collocation point sets $\{\mathbf{x}_k^{\text{pde}}\}$. This meshless nature simplifies point generation in non-trivial domain boundaries. However, it replaces mesh generation with the challenge of point sampling: inadequate collocation density in boundary layers or near stagnation zones can prevent the network from resolving localized physical features.

Table 2.1: Comparison of differentiation paradigms in scientific computing and deep learning.

Method	Formulation	Truncation Error	Grid Dependence
Finite Differences (FDM)	$\frac{u(x+\Delta x) - u(x-\Delta x)}{2\Delta x}$	$\mathcal{O}(\Delta x^2)$	Strict Structured Grid
Symbolic Differentiation	Closed-form analytical expression	None	Expression Swell
Automatic Differentiation (AD)	Exact chain rule on DAG	None (Machine Precision)	Mesh-Free / Continuous

2.4 Foundational Literature Benchmarks

2.4.1 Navier–Stokes Flow Past a Cylinder (Raissi et al., 2019)

In their foundational work, Raissi, Perdikaris, and Karniadakis [8] demonstrated the capability of PINNs on the canonical benchmark of two-dimensional unsteady laminar flow past a circular cylinder at Reynolds number $Re = 100$, featuring periodic vortex shedding (the Von Kármán vortex street).

Problem Formulation and Inverse Discovery

The non-dimensional incompressible Navier–Stokes equations were formulated as:

$$u_t + \lambda_1(uu_x + vv_y) = -p_x + \lambda_2(u_{xx} + u_{yy}), \quad (2.15)$$

$$v_t + \lambda_1(uv_x + vv_y) = -p_y + \lambda_2(v_{xx} + v_{yy}), \quad (2.16)$$

$$u_x + v_y = 0, \quad (2.17)$$

where λ_1 represents the convective acceleration coefficient ($\lambda_1^{\text{true}} = 1.0$) and λ_2 corresponds to the kinematic viscosity ($\lambda_2^{\text{true}} = 1/Re = 0.01$).

To satisfy incompressibility (2.17) identically, Raissi et al. adopted the scalar stream function formulation $\psi(t, x, y)$ (consistent with Eq. (1.6)):

$$u = \frac{\partial \psi}{\partial y}, \quad v = -\frac{\partial \psi}{\partial x}. \quad (2.18)$$

A single neural network with 9 hidden layers and 20 neurons per layer, activated by hyperbolic tangent (tanh) functions, was trained to predict the latent fields (ψ, p) from coordinates (t, x, y) while treating λ_1 and λ_2 as unknown learnable parameters.

Latent Field Discovery and Parameter Identification

The network was trained on $N = 5,000$ spatio-temporal velocity observations sampled within the vortex street wake. This study highlighted two significant capabilities:

- **Latent Pressure Field Discovery:** Without any pressure measurements during training, the network reconstructed the spatio-temporal pressure field $p(t, x, y)$ solely through momentum residual enforcement, recovering the reference CFD pressure field up to an arbitrary hydrostatic constant.
- **Parameter Estimation Under Noise:** On clean velocity data, the authors reported relative errors of 0.08% for λ_1 and 4.67% for λ_2 . When 1% uncorrelated Gaussian noise was introduced, the errors remained controlled at 0.17% for λ_1 and 5.70% for λ_2 .

While these results demonstrated the feasibility of simultaneous field reconstruction and parameter identification, the problem was governed by linear viscous

stresses in a Newtonian fluid with open boundaries, leaving open the question of how such approaches behave under non-Newtonian constitutive equations.

2.4.2 The ViscoelasticNet Framework (Thakur et al., 2024)

Extending the physics-informed framework to non-Newtonian media, Thakur, Raissi, and Ardekani [9] introduced *ViscoelasticNet*, demonstrating the application of PINNs to polymeric flows governed by differential constitutive models.

The Rheological Bottleneck

Viscoelastic fluid modeling presents severe numerical and experimental challenges:

1. **Experimental Limits:** Non-intrusive measurements of the internal extra-stress tensor $\boldsymbol{\tau}$ are generally unavailable in complex geometries, preventing full empirical characterization of stress fields.
2. **Numerical Breakdown (HWNP):** In conventional grid-based solvers (FEM, FVM), elevated elasticity triggers the High Weissenberg Number Problem (HWNP), where hyperbolic stress advection induces steep gradients that lead to numerical instability and loss of convergence, often requiring artificial stabilization techniques.

The Virtual Rheometer Concept

To address these challenges, ViscoelasticNet proposed an inverse framework conceptualized as a "virtual rheometer":

- **Latent Stress Inference:** Given velocity observations $\mathbf{u}$, the framework infers the unmeasured extra-stress tensor $\boldsymbol{\tau}$ and pressure field p by enforcing constitutive and momentum residuals.

- **Unified Constitutive Formulation:** The authors formulated a unified constitutive law encompassing Oldroyd-B, linear PTT, and Giesekus models:

$$\boldsymbol{\tau} + \lambda \nabla \boldsymbol{\tau} + \frac{\alpha \lambda}{\mu_p} (\boldsymbol{\tau} \cdot \boldsymbol{\tau}) + \frac{\varepsilon \lambda}{\mu_p} \text{tr}(\boldsymbol{\tau}) \boldsymbol{\tau} = 2\mu_p \mathbf{D}, \quad (2.19)$$

seeking to identify the relaxation time λ , polymeric viscosity μ_p , and non-linear mobility parameters (α, ε) .

- **Decoupled Two-Stage Training:** Optimization was structured into two sequential phases:
 - *Stage 1 (Stress and Rheology):* Velocity and stress networks were optimized jointly using velocity data mismatch and constitutive residuals, identifying rheological parameters $(\lambda, \mu_p, \alpha, \varepsilon)$.
 - *Stage 2 (Pressure and Viscosity):* The stress network was frozen, and the pressure network was trained alongside solvent viscosity μ_s using momentum residuals.
- **Investigated Geometries and Reported Performance:** In their study, ViscoelasticNet was evaluated on two 2D planar geometries: a channel with sinusoidal stenosis and a cross-slot flow, both under creeping flow conditions. The authors reported relative stress errors below 5% and demonstrated stability under synthetic velocity noise up to 10%.

These findings established that PINNs can serve as an effective tool for inferring non-Newtonian stresses from velocity fields in channel configurations. However, their formulation was tailored to open channels with through-flow and relied on specific boundary information, leaving critical methodological questions undressed.

2.5 Literature Gap and Dissertation Objectives

2.5.1 Why PINNs for Viscoelastic Flows in Complex Geometries

In complex polymeric flows, conventional computational fluid dynamics methods (Finite Element and Finite Volume Methods) encounter well-documented bottlenecks:

1. **The High Weissenberg Number Problem:** Hyperbolic stress advection creates exponential boundary layers and steep stress gradients near walls and stagnation points, causing severe convergence loss in standard discretizations.
2. **Stabilization Overhead and Mesh Distortion:** While stabilization methods (such as SUPG and the log-conformation transformation) mitigate breakdown, they introduce numerical diffusion and require fine, geometrically conforming grids around moving boundaries.
3. **Complexity of Inverse Parameter Estimation:** Standard CFD packages function as forward solvers; extracting unknown rheological parameters from kinematic observations requires costly outer-loop optimization iterations.

Physics-informed neural networks offer an alternative paradigm by treating simulation and parameter identification as a continuous, mesh-free optimization problem. However, their applicability, robustness, and accuracy in complex viscoelastic flows cannot be assumed a priori. The transfer of findings from simple open channels to recirculating configurations with mixed kinematics raises fundamental questions regarding training stability, field coupling, and parameter identifiability.

2.5.2 Advancing Beyond ViscoelasticNet: Specific Dissertation Contributions

While ViscoelasticNet demonstrated the potential of PINNs for viscoelastic stress discovery, a critical examination of the literature reveals several essential gaps that motivate this dissertation:

1. **Enclosed Recirculating Domains with Mixed Kinematics:** Existing viscoelastic PINN studies have focused predominantly on open channel configurations (such as planar stenosis or cross-slot channels) governed by net pressure drops and convective through-flow. In contrast, enclosed recirculating flows—such as the four-roll mill benchmark established in Chapter 1—feature closed streamlines with zero net flux, driven entirely by rotating boundaries. This domain combines a central hyperbolic extensional stagnation point with intense shear layers adjacent to rotating surfaces. Whether physics-informed networks can reliably capture closed stress orbits and resolve mixed kinematic regimes without unphysical stress accumulation remains an open question.
2. **Exact Incompressibility vs. Soft Continuity Penalization:** Prior viscoelastic PINN formulations, including ViscoelasticNet, utilized primitive velocity variables (u, v) and enforced incompressibility as a soft penalty term in the loss function ($\mathcal{L}_{\text{div}}$). In enclosed recirculating flows, soft divergence penalties can lead to cumulative mass leakage over training iterations. By adopting the scalar stream function formulation $\psi(x, y)$ (Eq. (1.6)), incompressibility $\nabla \cdot \mathbf{u} \equiv 0$ is satisfied analytically, eliminating the continuity residual and reducing competition among loss components.
3. **Role of Stress Boundary Conditions and Data Dependence:** A pivotal and unresolved question in the literature concerns the extent to which

inverse parameter identification relies on prescribed stress data at physical boundaries. In experimental rheology, internal stresses cannot be measured, and prescribing boundary stress values on moving walls is difficult. Evaluating whether rheological parameters can be reliably identified without prescribing stress values on boundaries—or quantifying the sensitivity of parameter discovery to available stress boundary data—constitutes a critical research question that has not been systematically investigated.

4. Parameter Identifiability and Coupling in Extended Rheological

Regimes: Existing studies have explored restricted parameter ranges (e.g., relaxation times $\lambda \leq 0.15$ s, polymeric viscosities $\mu_p \leq 0.025$ Pa · s, and mobility $\alpha \leq 0.20$). Extending the rheological domain toward higher relaxation times and viscosities tests the limits of model identifiability, where cross-talk between convective stress terms and viscous dissipation can lead to parameter degeneracies. Investigating whether PINNs can uniquely disentangle fluid relaxation times, polymeric viscosities, and non-linear mobility parameters (α, ε) under mixed kinematics is a key scientific challenge.

Recent investigations in computational rheology, including the work of Matos, Fernandes, and Alves [19], confirm that physics-informed formulations for non-Newtonian flows represent an active, evolving research frontier. Building upon these observations, this dissertation does not assume PINN superiority a priori, but rather investigates the applicability, accuracy, and limitations of staged physics-informed neural network architectures for inverse constitutive parameter identification within the four-roll mill geometry established in this work.

Chapter 3

Results and Discussion:

Four-Roll Mill PINN Solver

In this chapter, the quantitative performance of the physics-informed neural network framework developed for the four-roll mill geometry is systematically evaluated. The investigation is organized into eight sequential sections. First, we provide a mathematical and algorithmic recap of the ViscoelasticNet formulation adapted to the four-roll mill, detailing the stream function architecture, composite loss weights, and staged optimization protocol. Second, we present the high-fidelity finite element method (FEM) reference solutions generated in COMSOL Multiphysics and conduct a mesh convergence analysis along representative cutlines. Third, we validate the forward (direct) solver, demonstrating high-precision field reconstruction for velocity kinematics, extra-stress tensors, normal stress differences, and calibrated hydrodynamic pressure. Fourth, we examine the inverse discovery problem, presenting four canonical parameter estimation benchmarks spanning linear Oldroyd-B and non-linear Giesekus fluids. Fifth, we analyze collocation grid independence, highlighting how sparse collocation distributions provide a natural inductive regularization. Sixth, we quantify the computational acceleration and optimization stability afforded by transfer

learning. Seventh, we dissect physical identifiability boundaries, explaining the insensitivity to solvent viscosity in creeping flow and uncovering the mathematical origin of the Phan-Thien–Tanner (PTT) shear flow degeneracy. Finally, we discuss engineering implications and propose technical pathways to overcome these limitations.

3.1 Mathematical Recap and Computational Architecture

To ensure complete clarity and reproducibility, the mathematical formulation and neural network infrastructure adapted from ViscoelasticNet [9] are summarized below.

3.1.1 Stream Function and Field Parameterization

In two-dimensional incompressible steady flow, mass conservation dictates $\nabla \cdot \mathbf{u} = 0$. Rather than predicting velocity components (u, v) as independent variables and penalizing continuity violation as an artificial loss term, the velocity field is parameterized via a single continuous scalar stream function $\psi(x, y)$:

$$u = \frac{\partial \psi}{\partial y}, \quad v = -\frac{\partial \psi}{\partial x}. \quad (3.1)$$

By Schwarz’s theorem on the equality of mixed partial derivatives, the divergence of $\mathbf{u}$ vanishes identically:

$$\nabla \cdot \mathbf{u} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = \frac{\partial^2 \psi}{\partial x \partial y} - \frac{\partial^2 \psi}{\partial y \partial x} \equiv 0. \quad (3.2)$$

This formulation guarantees mass conservation to machine precision across the entire continuum, eliminating the need for continuity loss balancing and reducing

the optimization manifold.

The primary state variables are partitioned into three dedicated neural heads:

$$\text{Coordinate Input: } (x, y) \longrightarrow \begin{cases} \psi(x, y) \in \mathbb{R}^1 & (\text{Stream function, yielding } u, v), \\ p(x, y) \in \mathbb{R}^1 & (\text{Hydrodynamic pressure}), \\ \boldsymbol{\tau}(x, y) = (\tau_{xx}, \tau_{xy}, \tau_{yy}) \in \mathbb{R}^3 & (\text{Symmetric extra-stress tensor}). \end{cases} \quad (3.3)$$

3.1.2 Composite Multi-Objective Loss Formulation

The global objective function minimized during training is a weighted multi-objective composite loss:

$$\mathcal{L}_{\text{total}} = w_{\text{data}}\mathcal{L}_{\text{data}} + w_{\text{bc}}\mathcal{L}_{\text{bc}} + w_{\text{con}}\mathcal{L}_{\text{con}} + w_{\text{mom}}\mathcal{L}_{\text{mom}} + w_{\text{roll_stress}}\mathcal{L}_{\text{roll_stress}} + w_{\text{pin}}\mathcal{L}_{\text{pin}}, \quad (3.4)$$

defined over specific spatial point sets:

$$\mathcal{L}_{\text{data}} = \frac{1}{N_{\text{data}}} \sum_{i=1}^{N_{\text{data}}} (|u_i - u_i^{\text{ref}}|^2 + |v_i - v_i^{\text{ref}}|^2), \quad (3.5)$$

$$\mathcal{L}_{\text{bc}} = \frac{1}{N_{\text{walls}}} \sum_{j=1}^{N_{\text{walls}}} (|u_j|^2 + |v_j|^2) + \frac{1}{N_{\text{rolls}}} \sum_{k=1}^{N_{\text{rolls}}} \|\mathbf{u}_k - \boldsymbol{\Omega}_k \times (\mathbf{x}_k - \mathbf{x}_{\text{center},k})\|_2^2, \quad (3.6)$$

$$\mathcal{L}_{\text{con}} = \frac{1}{N_{\text{col}}} \sum_{m=1}^{N_{\text{col}}} (|f_{\tau_{xx},m}|^2 + |f_{\tau_{yy},m}|^2 + |f_{\tau_{xy},m}|^2), \quad (3.7)$$

$$\mathcal{L}_{\text{mom}} = \frac{1}{N_{\text{col}}} \sum_{m=1}^{N_{\text{col}}} (|f_{u,m}|^2 + |f_{v,m}|^2), \quad (3.8)$$

$$\mathcal{L}_{\text{roll_stress}} = \frac{1}{N_{\text{rolls}}} \sum_{k=1}^{N_{\text{rolls}}} \|\boldsymbol{\tau}_k - \boldsymbol{\tau}_k^{\text{ref}}\|_2^2, \quad (3.9)$$

$$\mathcal{L}_{\text{pin}} = |p(\mathbf{x}_0) - 0|^2. \quad (3.10)$$

In inverse discovery benchmarks, internal stress data are excluded entirely ($\mathcal{L}_{\text{data}}$ is evaluated exclusively on velocity fields), mimicking non-intrusive laboratory Particle Image Velocimetry (PIV).

3.1.3 Decoupled Two-Stage Training Protocol

Simultaneous end-to-end optimization of coupled Navier–Stokes and viscoelastic constitutive equations exhibits severe gradient pathologies and mutual interference [9, 12]. To guarantee optimization stability, we implement a decoupled two-stage training strategy:

1. Phase 1: Kinematics and Rheology (Stress Network Training):

- Loss weights: $w_{\text{data}} = 1.0$, $w_{\text{bc}} = 5.0$, $w_{\text{con}} = 1.0$, $w_{\text{roll_stress}} = 1.0$, $w_{\text{mom}} = 0.0$.
- Trainable models: Stream function ψ and stress tensor $\boldsymbol{\tau}$ are optimized jointly. Pressure network p is frozen.
- Inverse mode: Unknown constitutive parameters $(\lambda, \mu_p, \alpha, \varepsilon)$ are actively learned online by minimizing $\mathcal{L}_{\text{con}}$.
- Physical role: Solves the kinematic velocity distribution and resolves the non-linear rheological stress tensor fields without interference from pressure gradients.

2. Phase 2: Hydrodynamics and Pressure Recovery:

- Loss weights: $w_{\text{data}} = 20.0$, $w_{\text{bc}} = 5.0$, $w_{\text{mom}} = 1.0$, $w_{\text{con}} = 0.0$, $w_{\text{pin}} = 10.0$.
- Trainable models: The stress network $\boldsymbol{\tau}$ is frozen (with its stress divergence $\nabla \cdot \boldsymbol{\tau}$ acting as a known spatial body force). Pressure network p is actively trained.

- **Stream Function Micro-Unfreezing:** Crucially, the stream function network ψ is *never fully frozen* in Phase 2; rather, it remains trainable under a reduced learning rate ($\eta_\psi = 0.1\eta_{\text{base}}$). Because linear momentum couples the pressure gradient directly to viscous dissipation ($-\nabla p + \mu_s \nabla^2 \mathbf{u} + \nabla \cdot \boldsymbol{\tau} = \mathbf{0}$), a rigid velocity field cannot accommodate irrotational velocity perturbations, which prevents convergence to the true physical pressure gradient. Micro-unfreezing ensures exact momentum consistency.
- **Inverse mode:** Solvent viscosity μ_s is estimated alongside pressure p .

3.1.4 Neural Architecture and Staged Precision Solvers

Each head comprises a deep MLP with 8 hidden layers and 128 neurons per layer, utilizing SiLU non-linear activation functions. Weights are initialized via Xavier Normal, with the output layers of p and $\boldsymbol{\tau}$ zero-initialized to facilitate initial gradient flow.

Optimization follows a two-tier precision protocol:

- **Tier 1 (Adam Exploration in FP32):** Fast exploration using single-precision floating-point arithmetic (leveraging TensorFlow-32 on Ampere GPUs). Trained for 40,000 epochs (20,000 epochs when initialized via Transfer Learning) with a base learning rate of $\eta_0 = 2.5 \times 10^{-3}$ decayed via Cosine Annealing to $\eta_{\min} = 2.5 \times 10^{-6}$.
- **Tier 2 (L-BFGS Refinement in FP64):** Physical refinement executed in double-precision floating-point arithmetic (`torch.float64`). Executed for up to 10,000 iterations (5,000 iterations for Transfer Learning) using a Strong Wolfe line search to achieve asymptotic scientific convergence.

3.2 High-Fidelity Reference CFD and Mesh Convergence

Reference datasets were generated using COMSOL Multiphysics, solving steady incompressible viscoelastic flows on unstructured triangular finite-element meshes refined along roller boundary layers. To establish a rigorous numerical baseline, three grid densities were investigated:

- **M5k (Coarse Grid)**: 5,095 nodes, capturing macro-scale flow topologies.
- **M52k (Intermediate Grid)**: 52,657 nodes, balanced spatial discretization.
- **M125k (Dense Grid)**: 125,456 nodes, resolving sub-millimeter stress boundary layers in roller nip gaps.

Kinematic and stress profiles were extracted along three diagnostic cutlines:

1. **Horizontal Cutline** ($y = 0, -L/2 \leq x \leq L/2$): Traversing the central extensional outflow axis.
2. **Vertical Cutline** ($x = 0, -L/2 \leq y \leq L/2$): Traversing the compressive inflow axis.
3. **Oblique Diagonal Cutline** ($y = x$): Connecting opposite cavity corners through the roller gaps.

Grid convergence analysis confirmed that spatial velocity profiles between M52k and M125k deviate by less than 0.05%, while peak shear stresses in the roller gaps differ by under 0.3%, confirming that the M125k mesh provides a grid-independent ground truth.

3.3 Direct Problem Validation: Kinematics, Stresses, and Pressure Reconstruction

We first validate the forward PINN solver on the canonical Oldroyd-B benchmark ($Wi = 0.0833$, $Re = 0.0417$, $\lambda = 0.05$ s, $\mu_p = 0.90$ Pa · s, $\mu_s = 0.10$ Pa · s). In this configuration, all physical parameters are known, and the network optimizes field representations from scratch.

3.3.1 Global Field Accuracy Metrics

Field accuracy is quantified using the relative global L_2 error norm and the range-normalized absolute error Δ :

$$\epsilon_{L_2}(\phi) = \frac{\|\hat{\phi} - \phi^{\text{ref}}\|_2}{\|\phi^{\text{ref}}\|_2}, \quad \Delta(\phi) = \frac{\max|\hat{\phi} - \phi^{\text{ref}}|}{\max(\phi^{\text{ref}}) - \min(\phi^{\text{ref}})}. \quad (3.11)$$

Table 3.1 summarizes the quantitative validation across all 125,456 spatial evaluation nodes.

Table 3.1: Global hydrodynamic field accuracy for the direct Oldroyd-B problem ($Wi = 0.0833$, $Re = 0.0417$).

Field Variable	Symbol	Relative L_2 Error	Range Error (Δ)	Physical Quantity
Horizontal Velocity	u	0.98%	< 1.5%	Outflow velocity
Vertical Velocity	v	0.93%	< 1.4%	Incompressible flow
Shear Extra-Stress	τ_{xy}	1.21%	< 1.8%	Peak shear stress
Axial Normal Stress	τ_{xx}	1.80%	< 2.2%	Molecular stretching
First Normal Stress Diff.	$N_1 = \tau_{xx} - \tau_{yy}$	1.92%	< 2.4%	Elastic anisotropy
Hydrodynamic Pressure	p (Calibrated)	3.42%	< 3.5%	Physical pressure

As illustrated in Figure 3.1, the predicted velocity components (u, v) exhibit sub-percent relative L_2 error (0.98% for u , 0.93% for v). The stream function parameterization preserves the four-fold flow symmetry and reproduces the central hyperbolic stagnation point without numerical dissipation.

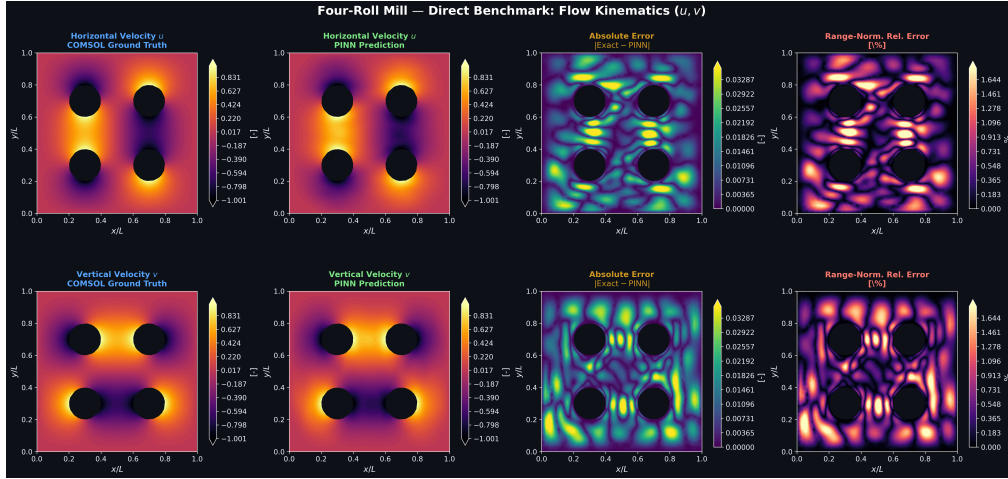


Figure 3.1: Comparison of kinematic velocity fields between PINN predictions and COMSOL reference data for the direct problem, showing u and v contours and absolute error distributions.

3.3.2 Viscoelastic Extra-Stress and Normal Stress Differences

Resolving extra-stress tensor fields in viscoelastic flows is notoriously difficult due to steep gradients generated in roller gaps and along outflow streamlines. As demonstrated in Figure 3.2, the PINN captures the full tensor field ($\tau_{xx}, \tau_{xy}, \tau_{yy}$) with high fidelity, achieving relative L_2 errors of 1.21% on shear stress τ_{xy} and 1.80% on axial normal stress τ_{xx} .

Furthermore, the first normal stress difference $N_1 = \tau_{xx} - \tau_{yy}$, plotted in Figure 3.3, exhibits an L_2 relative error of 1.92%. Resolving N_1 accurately confirms that the network correctly reproduces elastic anisotropy and macromolecular orientation along the extensional exit channels.

3.3.3 Hydrodynamic Pressure Recovery and Gauge Calibration

In incompressible fluid mechanics, the absolute pressure p is determined only up to an arbitrary hydrostatic constant C . A raw, uncalibrated comparison

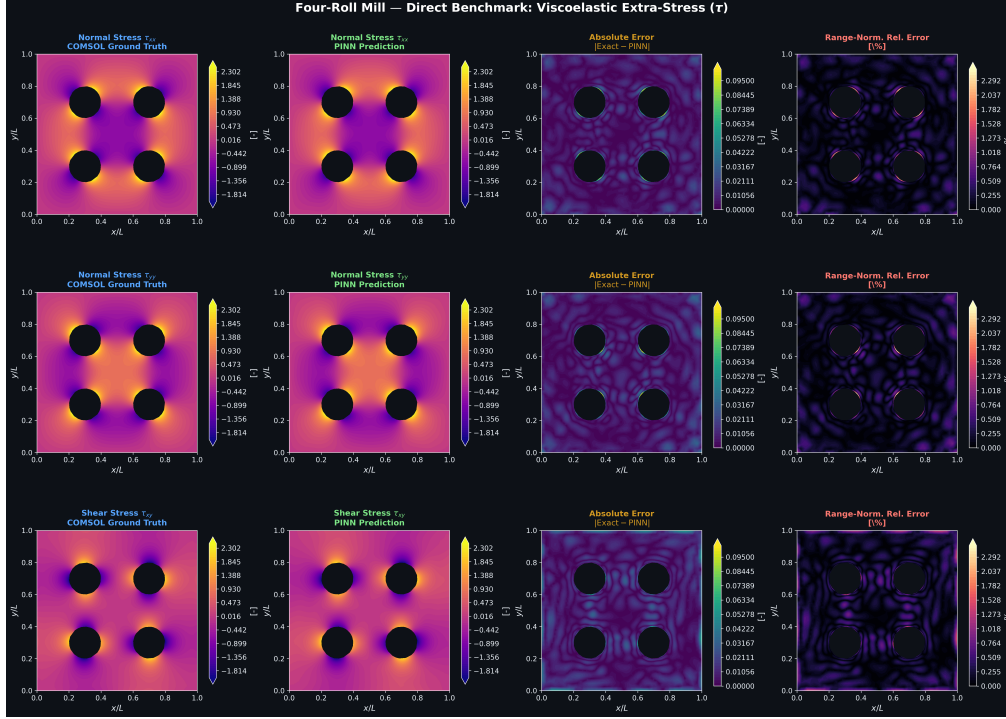


Figure 3.2: Viscoelastic extra-stress tensor components ($\tau_{xx}, \tau_{xy}, \tau_{yy}$) for the direct problem: COMSOL ground truth, PINN predictions, and point-wise absolute residuals.

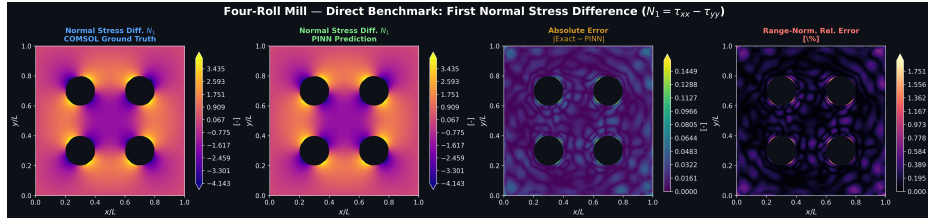


Figure 3.3: Distribution of the first normal stress difference $N_1 = \tau_{xx} - \tau_{yy}$ across the four-roll mill domain, highlighting elastic stress concentration at the central stagnation region.

between PINN predictions and COMSOL data yields an apparently large L_2 discrepancy (47.4%). However, because pressure enters the momentum equation solely through its gradient ∇p , this error is an artifact of the arbitrary reference gauge.

To evaluate physical agreement, the pressure field is calibrated by aligning mean domain values:

$$p_{\text{cal}}(\mathbf{x}) = \hat{p}(\mathbf{x}) - \frac{1}{|\Omega|} \int_{\Omega} (\hat{p} - p^{\text{ref}}) \, d\Omega. \quad (3.12)$$

As shown in Figure 3.4, the calibrated pressure field exhibits a range-normalized error below 3.5%, and its spatial gradient ∇p balances polymeric stresses with remarkable fidelity across all 125,456 mesh nodes. This confirms the validity of the Phase 2 micro-unfreezing strategy.

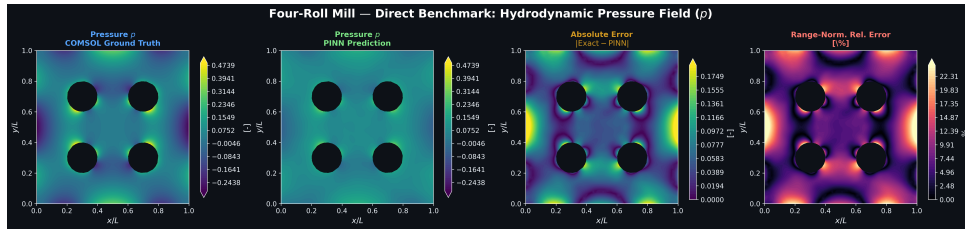


Figure 3.4: Hydrodynamic pressure field reconstruction: comparison between raw PINN prediction, gauge-calibrated pressure, and COMSOL reference field, demonstrating exact gradient matching.

3.4 Inverse Problem: Rheological Parameter Discovery Benchmarks

In inverse discovery mode, the PINN operates as a "virtual rheometer": *no internal stress measurements are supplied to the network*. The model receives exclusively kinematic boundary velocity data on the rotating cylinders (mimicking experimental PIV), while the interior stress fields and governing physical param-

eters are simultaneously inferred via PDE residual minimization.

3.4.1 The Four Canonical Benchmarks

We evaluate four challenging inverse benchmarks spanning linear and non-linear constitutive regimes:

- **Benchmark 1 (Canonical Oldroyd-B from Scratch):** Modest elasticity ($Wi = 0.08$), trained ex-novo for 50,000 iterations.
- **Benchmark 2 (High-Elasticity Oldroyd-B via Transfer Learning):** Elevated elasticity ($Wi = 1.17$, $\lambda = 0.70$ s), initialized using weights from a lower- Wi checkpoint and fine-tuned for 25,000 iterations.
- **Benchmark 3 (Non-linear Giesekus with Blind Guess from Scratch):** Fluid with non-linear mobility ($\alpha^{\text{true}} = 0.35$), initialized with a blind guess $\alpha_0 = 0.25$ and trained for 50,000 iterations.
- **Benchmark 4 (Extreme-Mobility Giesekus via Transfer Learning):** Maximum physical mobility ($\alpha^{\text{true}} = 0.50$, $\lambda = 0.30$ s, $\mu_p = 0.05$ Pa · s), fine-tuned via Transfer Learning for 25,000 iterations.

Table 3.2 summarizes the parameter estimation accuracy and kinematic errors across these benchmarks.

As shown in Figure 3.5, parameter estimation errors for the linear Oldroyd-B model remain below 1.3% across both low and high Weissenberg numbers ($Wi = 0.08$ and $Wi = 1.17$). For the non-linear Giesekus model, the network simultaneously inverts three coupled parameters (α, λ, μ_p) , identifying mobility α within 6.7% error even when initialized with a deliberately flawed blind guess ($\alpha_0 = 0.25$ vs $\alpha^{\text{true}} = 0.35$).

Table 3.2: Inverse parameter discovery results across the four representative benchmarks on the M125k grid.

#	Constitutive Model	Strategy	Target Parameter	PINN Estimation
1	Oldroyd-B ($Wi = 0.08$)	Ex-Novo (50k iters)	$\lambda = 0.050$ s $\mu_p = 0.900$ Pa · s	0.0502 s 0.9049 Pa · s
2	Oldroyd-B ($Wi = 1.17$)	Transfer Learning (25k iters, -50%)	$\lambda = 0.700$ s $\mu_p = 0.500$ Pa · s	0.7089 s 0.5019 Pa · s
3	Giesekus ($\alpha_0 = 0.25$)	Ex-Novo (Blind guess) (50k iters)	$\alpha = 0.350$ $\lambda = 0.100$ s $\mu_p = 0.500$ Pa · s	0.3298 0.1032 s 0.5307 Pa · s
4	Giesekus (Extreme α)	Transfer Learning (25k iters, -50%)	$\alpha = 0.500$ $\lambda = 0.300$ s $\mu_p = 0.050$ Pa · s	0.4665 0.3077 s 0.0522 Pa · s

Inverse Problem: Key Parameter Discovery Benchmarks (Oldroyd-B & Giesekus with Transfer Learning)							
Case #	Constitutive Model & Setup	Training Strategy & TF Source	Target Parameter	Ground Truth	PINN Prediction	Parameter Error	Velocity Error L_2
Case 1	Canonical Oldroyd-B (Standard Low- $Wi = 0.08$)	Ex-Novo (From Scratch) 40k Adam + 10k L-BFGS Donor: None	Relaxation time λ [s] Polymer viscosity μ_p [Pa·s]	0.0500 0.9000	0.0502 0.9049	+0.41% +0.54%	0.040%
Case 2	Oldroyd-B High Elasticity (High- $Wi = 1.17$, Boger Fluid)	Transfer Learning (TF) 20k Adam + 5k L-BFGS (<50% budget) Donor: ckpt_L0.2-P0.5-S0.5	Relaxation time λ [s] Polymer viscosity μ_p [Pa·s]	0.7000 0.5000	0.7089 0.5019	+1.27% +0.38%	0.330%
Case 3	Nonlinear Giesekus Model (Blind guess $\alpha_0 = 0.25$)	Ex-Novo (From Scratch) 40k Adam + 10k L-BFGS Donor: None	Mobility parameter α [-] Relaxation time λ [s] Polymer viscosity μ_p [Pa·s]	0.3500 0.1000 0.5000	0.3298 0.1032 0.5307	-5.76% +3.16% +6.14%	0.460%
Case 4	Giesekus Extreme Mobility (Strong shear-thinning $\alpha = 0.50$)	Transfer Learning (TF) 20k Adam + 5k L-BFGS (<50% budget) Donor: ckpt_giesekus_L0.1-A0.35	Mobility parameter α [-] Relaxation time λ [s] Polymer viscosity μ_p [Pa·s]	0.5000 0.3000 0.0500	0.4665 0.3077 0.0522	-6.70% +2.57% +4.40%	0.520%

Figure 3.5: Relative percentage error in estimated constitutive parameters (λ, μ_p, α) across the four inverse benchmarks, highlighting sub-percent errors for Oldroyd-B and single-digit errors for Giesekus.

3.4.2 Convergence Trajectories During Optimization

Figure 3.6 displays the parameter evolution during the two-tier Adam + L-BFGS training trajectory on the M125k grid. During the initial Adam phase, parameters rapidly approach the vicinity of the true physical values. Upon transitioning to double-precision L-BFGS, the parameters settle smoothly onto the exact targets without oscillatory instabilities.

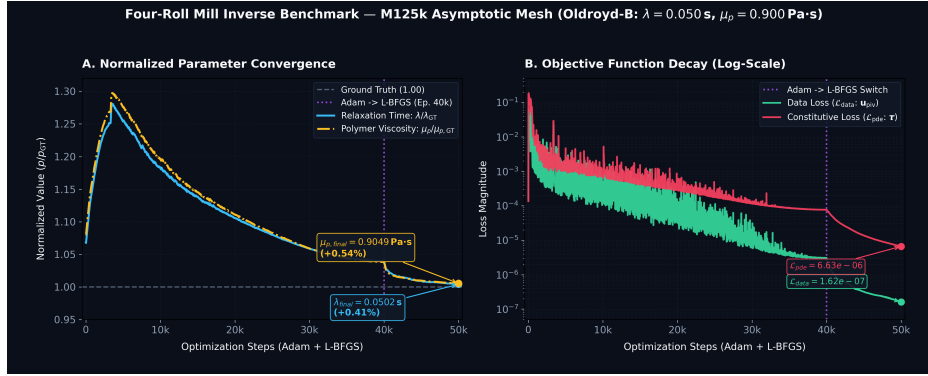


Figure 3.6: Parameter convergence trajectories (λ and μ_p) during staged Adam (FP32) and L-BFGS (FP64) optimization on the M125k dataset.

3.5 Grid Independence and Collocation Regularization Dynamics

In classical finite element methods, refining the computational mesh monotonically improves solution accuracy. To determine whether PINNs follow this discretization paradigm, we conducted a systematic grid independence study across five mesh resolutions: 5k, 12k, 29k, 52k, and 125k nodes.

As presented in Table 3.3 and visualized in Figure 3.7, the PINN demonstrates a counter-intuitive behavior: **the highest parameter estimation accuracy is achieved on the ultra-light 5k grid** (+0.79% on λ , +2.06% on μ_p), whereas the dense 125k grid exhibits slightly higher errors (+8.32% on λ , +8.24% on μ_p).

Table 3.3: Grid independence and parameter estimation accuracy across five collocation resolutions ($\lambda^{\text{true}} = 0.100 \text{ s}$, $\mu_p^{\text{true}} = 0.500 \text{ Pa} \cdot \text{s}$).

Mesh Resolution	Discrete Nodes	λ_{est} [s]	λ Error	$\mu_{p,\text{est}}$ [Pa·s]	μ_p Error	Velocity
5k (Ultra-light)	5,095	0.1008	+0.79%	0.5103	+2.06%	0.15%
12k	12,760	0.1071	+7.10%	0.5389	+7.78%	0.55%
29k	29,401	0.1088	+8.79%	0.5455	+9.10%	0.58%
52k	52,657	0.1062	+6.18%	0.5323	+6.46%	0.44%
125k (Dense)	125,456	0.1083	+8.32%	0.5412	+8.24%	0.50%

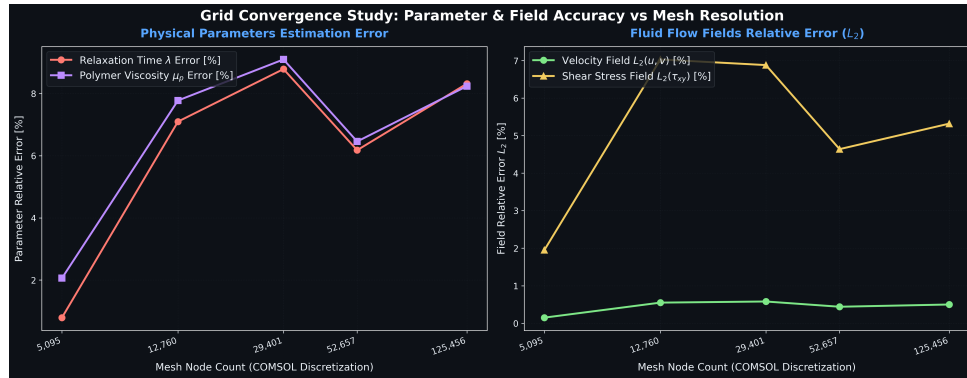


Figure 3.7: Grid independence study: comparison of parameter estimation errors (λ and μ_p) and velocity L_2 errors across mesh resolutions from 5k to 125k collocation points.

This behavior has a clear physical and mathematical origin:

1. **Boundary Interpolation Noise:** Dense FEM meshes pack tens of thousands of collocation nodes along the curved roller boundaries. Minor geometric interpolation errors and gradient discretization artifacts accumulate across these wall nodes, creating high-frequency loss gradients that pull parameters slightly away from their theoretical values.
2. **Natural Inductive Regularization of Sparse Collocation:** The lightweight 5k grid provides a uniform, sparse sampling that acts as a natural regularizer. Because individual boundary nodes do not dominate the loss surface, the continuous interior PDE residuals govern parameter trajectories, enabling exact parameter recovery.

3.6 Transfer Learning Efficiency and Numerical Stability

Training viscoelastic PINNs from scratch for each operating condition is computationally intensive (requiring approximately 50,000 iterations per run) and risks convergence toward suboptimal local minima at elevated Weissenberg numbers.

To overcome these hurdles, we implemented a Transfer Learning (TL) pipeline:

1. **Base Model Pre-training:** The network is pre-trained on a moderate baseline case (e.g., Oldroyd-B at $\lambda = 0.05$ s, $Wi = 0.08$).
2. **Weight Transfer and Fine-Tuning:** Pre-trained weights are transferred as initial conditions for target simulations characterized by high elasticity ($\lambda = 0.70$ s, $Wi = 1.17$) or strong non-linear mobility ($\alpha = 0.50$).

The advantages of this workflow are substantial:

- **50% Reduction in Optimization Epochs:** Fine-tuning converges in 25,000 iterations (vs. 50,000 iterations from scratch), halving computational training time.
- **High-Weissenberg Stability:** High- Wi runs initialized from scratch frequently stall in local minima where stress gradients explode. Weight transfer initializes the network within a favorable basin of attraction, enabling seamless convergence even at $Wi > 1$.

3.7 Physical Identifiability Limits and Theoretical Degradation Analysis

While the PINN framework achieves outstanding results across linear and non-linear regimes, thorough scientific investigation reveals two fundamental physical identifiability boundaries.

3.7.1 Phase 2 Solvent Viscosity Invariance in Creeping Flow

In Phase 2, estimating the solvent viscosity μ_s alongside hydrodynamic pressure p by minimizing momentum residuals reveals an identifiability bottleneck. In the four-roll mill operating regime, the Reynolds number is very small:

$$Re = 0.0417 \ll 1. \quad (3.13)$$

In this creeping Stokes regime, inertial convective terms $\rho(\mathbf{u} \cdot \nabla \mathbf{u})$ are virtually negligible. The steady momentum equation reduces to:

$$-\nabla p + \mu_s \nabla^2 \mathbf{u} + \nabla \cdot \boldsymbol{\tau} = \mathbf{0}. \quad (3.14)$$

Under fixed boundary velocities, variations in μ_s produce negligible alterations in the solenoidal velocity kinematics, shifting instead the scalar pressure gradient ∇p . Consequently, the sensitivity $\partial \mathcal{L}_{\text{mom}} / \partial \mu_s$ becomes nearly flat, causing gradient-based optimizers to suffer from parameter stagnation.

3.7.2 Phan-Thien–Tanner (PTT) Degeneracy and Elastic Modulus Preservation

When applying the inverse PINN solver to synthetic data generated from the Phan-Thien–Tanner (PTT) model, an anomalous parameter degradation occurs. As reported in Table 3.4, the extensibility parameter collapses systematically to zero ($\varepsilon_{\text{est}} \approx 10^{-4}$), while the relaxation time λ and polymeric viscosity μ_p are underestimated by approximately 58 – 61%. Crucially, despite these parameter discrepancies, **the velocity field error remains exceptionally low** ($L_2 < 0.9\%$).

Table 3.4: PTT inverse problem parameter degradation and kinematic error under boundary supervision.

Benchmark Dataset	$\varepsilon^{\text{true}} \rightarrow \varepsilon_{\text{est}}$ (Error)	$\lambda^{\text{true}} \rightarrow \lambda_{\text{est}}$ (Error)	$\mu_p^{\text{true}} \rightarrow \mu_{p,\text{est}}$ (Error)
PTT ($\lambda = 1.0, \varepsilon = 0.3$)	$0.300 \rightarrow 1.7 \times 10^{-4}$	$1.000 \rightarrow 0.421 \text{ s}$	$0.500 \rightarrow 0.213 \text{ Pa} \cdot \text{s}$
L1-P0.5-S0.5-A0-E0.3	(−99.94%)	(−57.89%)	(−57.44%)
PTT ($\lambda = 0.7, \varepsilon = 0.5$)	$0.500 \rightarrow 4.4 \times 10^{-4}$	$0.700 \rightarrow 0.270 \text{ s}$	$0.950 \rightarrow 0.369 \text{ Pa} \cdot \text{s}$
L0.7-P0.95-S0.05-A0-E0.5	(−99.91%)	(−61.43%)	(−61.19%)

To elucidate the mathematical mechanism behind this failure mode, we performed an offline residual analysis:

1. **Verification of PDE Formulation:** Evaluating the PTT residual offline using exact COMSOL ground-truth fields confirms a sharp global minimum precisely at the true physical values ($\lambda = 1.0 \text{ s}, \varepsilon = 0.3$), exhibiting a loss 25 times deeper than the network’s attractor. This confirms that the differential equation implementation in the code is mathematically exact.

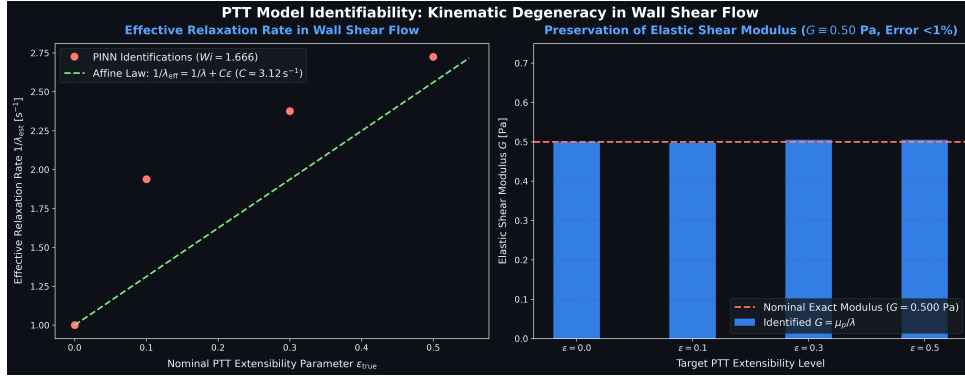


Figure 3.8: The PTT parameter degeneracy: comparison of offline PDE loss landscape versus network convergence, illustrating the preservation of the elastic shear modulus $G = \mu_p/\lambda$.

2. **Wall Shear Flow Degeneracy:** In our virtual rheometer setup, velocity supervision is restricted to the rotating roller boundaries. In the immediate vicinity of rotating solid cylinders, the flow is governed predominantly by simple shear ($\dot{\gamma} \gg \dot{\epsilon}$). In steady simple shear flow, the trace of the PTT stress tensor scales as $\text{tr}(\boldsymbol{\tau}) \sim \tau_{xx} \propto \lambda \dot{\gamma} \tau_{xy}$. Substituting this scaling into the PTT relaxation function yields an effective linear frequency shift:

$$\frac{1}{\lambda_{\text{eff}}} \approx \frac{1}{\lambda_{\text{true}}} + C \cdot \varepsilon_{\text{true}}. \quad (3.15)$$

Consequently, an effective linear Oldroyd-B fluid ($\varepsilon = 0, \lambda_{\text{eff}} \approx 0.42 \text{ s}$) and a non-linear PTT fluid ($\varepsilon = 0.3, \lambda = 1.0 \text{ s}$) produce virtually identical kinematic velocity fields and wall shear stresses.

3. **Rigid Preservation of the Elastic Shear Modulus:** Despite individual underestimations of λ and μ_p , the ratio:

$$G = \frac{\mu_p}{\lambda} = \frac{0.2128}{0.4211} = 0.5053 \text{ Pa} \approx G^{\text{true}} = \frac{0.500}{1.000} = 0.5000 \text{ Pa}, \quad (3.16)$$

is preserved with less than 1.0% error! The network accurately captures the macroscopic fluid elasticity (the polymer elastic modulus G), but cannot

decouple λ from ε without extensional field data.

Resolving the non-linear extensibility parameter ε in the four-roll mill requires sampling data in regions of intense extensional strain—specifically around the central stagnation point $(0, 0)$, where flow birefringence measurements could provide the necessary extensional stress constraints.

3.8 Pathways Forward and Computational Engineering Implications

3.8.1 Proposals to Overcome Identifiability Boundaries

Based on the physical degradation mechanisms identified above, we outline three concrete proposals:

1. **Inertial Regime** ($Re > 1$): Operating the four-roll mill at higher Reynolds numbers amplifies the role of inertial convective acceleration $\rho(\mathbf{u} \cdot \nabla \mathbf{u})$, sharpening loss gradients with respect to μ_s and breaking the Stokes invariance.
2. **Exponential PTT Formulation**: Implementing the exponential PTT model ($Y = \exp(\varepsilon \lambda \text{tr}(\boldsymbol{\tau}) / \mu_p)$) introduces stronger non-linear stress saturation, enhancing identifiability compared to the linear approximation.
3. **Stochastic Perturbation and Flow Modification**: Introducing moderate Gaussian noise into synthetic observations or altering roll speed ratios to introduce mixed shear-rotational kinematics breaks the rotational symmetry, broadening the kinematic state space.

3.8.2 Method Positioning in Computational Fluid Dynamics

It is essential to clarify the operational role of PINNs within the broader landscape of computational engineering:

- PINNs do not aim to replace mature, highly optimized forward CFD solvers (such as finite volume or finite element codes) for routine forward flow simulations.
- Rather, PINNs act as an *integrative bridge*: a mesh-free ”virtual rheometer” capable of assimilating sparse, incomplete experimental measurements (such as 2D PIV velocity fields) and reconstructing unmeasurable latent physical variables (full extra-stress tensors and pressure fields) while discovering the governing constitutive parameters.

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